

Verification and Validation Report: Software Engineering

Team #12, Streamliners

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1 Revision History

Date	Version	Notes
Date 1	1.0	Notes
Date 2	1.1	Notes

2 Symbols, Abbreviations and Acronyms

Symbol	Description
T	Test

[symbols, abbreviations or acronyms – you can reference the SRS tables if needed —SS]

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3 Functional Requirements Evaluation

4 Nonfunctional Requirements Evaluation

4.1 Usability

4.2 Performance

4.3 etc.

5 Comparison to Existing Implementation

This section will not be appropriate for every project.

6 Unit Testing

6.1 Event Creation and Management Module

Create Event Screen

Test ID	Inputs	Expected Values	Actual Values	Result
ECM1	Open the create event screen.	The screen displays the event form, input fields, toggles, and Save/Cancel buttons.	The screen displayed all required form controls correctly.	Pass
ECM2	Press Save with required fields left empty.	Event creation is blocked and an error message is shown for missing required fields.	Submission was blocked and an error alert was shown.	Pass
ECM3	Enter valid event data and press Save.	The event is created successfully, price is converted to cents, and the user is redirected to the confirmation page.	The event was created with the correct payload and navigation occurred.	Pass
ECM4	Enter valid data but simulate API failure on Save.	Event creation fails gracefully and an error alert is shown.	The API error was caught and the correct alert was displayed.	Pass
ECM5	Press Cancel on the create event screen.	The user is returned to the previous screen without creating an event.	The screen navigated back successfully.	Pass

Events Controller / Service

Test ID	Inputs	Expected Values	Actual Values	Result
ECM6	Request all events from the backend.	The system returns the list of stored events.	All events were returned correctly.	Pass
ECM7	Request one event by id.	The correct event is returned if it exists; otherwise null is returned.	Existing events were returned correctly and missing events returned null.	Pass
ECM8	Create an event requiring bus or table signup.	The event is created and corresponding seat records are initialized automatically.	Event creation and seat generation both succeeded.	Pass
ECM9	Update or delete an existing event.	The requested event is modified or deleted successfully.	Update and delete operations completed correctly.	Pass

6.2 Ticket Purchase and Payment System Module

Events Screen / Ticket Access

Test ID	Inputs	Expected Values	Actual Values	Result
TP1	Load an available event with price and registration count.	The event card displays the correct event name, location, price, and open status.	The event card showed the correct information.	Pass
TP2	Load an event that has reached capacity.	The event is marked Full and signup is disabled.	Full badge and disabled action were shown correctly.	Pass
TP3	Load a past event.	The event is marked Past and signup is disabled.	Past badge and ended status were shown correctly.	Pass
TP4	Click Sign Up on an available event.	The user is routed to the event signup page for that event.	Navigation to the signup page occurred successfully.	Pass
TP5	Load events for a user who is already registered.	The system shows the registered status for the user's events.	Registered events were correctly marked as signed up.	Pass

Ticket Validation / Check-In

Test ID	Inputs	Expected Values	Actual Values	Result
TP6	Submit empty or invalid QR code data.	The system rejects the QR code and returns an invalid QR error.	Invalid QR codes were rejected correctly.	Pass
TP7	Submit a valid QR code for a ticket that does not exist.	The system returns a ticket not found error.	Missing tickets were handled correctly.	Pass
TP8	Submit a valid QR code for a ticket already checked in.	The system returns an already checked-in response without updating the record.	Already checked-in tickets were detected correctly.	Pass
TP9	Submit a valid QR code for a valid unchecked ticket.	The ticket is marked as checked in and success is returned.	The ticket was updated and validated successfully.	Pass

6.3 Bus and Table Registration Module

Event Signup / Seat Initialization

Test ID	Inputs	Expected Values	Actual Values	Result
BTR1	Create an event with table signup enabled and a valid table configuration.	Table seat records are automatically created for the event.	Table seats were generated correctly.	Pass
BTR2	Create an event with bus signup enabled and a valid bus configuration.	Bus seat records are automatically created for the event.	Bus seats were generated correctly.	Pass
BTR3	Sign up a user for an event with an optional table selection.	The selected table information is passed correctly to the backend during signup.	The signup call included the correct event, user, and table id.	Pass
BTR4	Cancel an existing event signup.	The user's signup is removed successfully.	The signup was cancelled correctly.	Pass
BTR5	Attempt to cancel signup when no signup exists.	The system returns an appropriate error and does not modify data.	The system returned the expected error response.	Pass

6.4 Role-Based Access Control (RBAC) Module

Frontend and Backend Authorization

Test ID	Inputs	Expected Values	Actual Values	Result
RBAC1	Load the events screen as an admin user.	The Create Event button is visible and can be used.	The admin user could see and access event creation.	Pass
RBAC2	Load the events screen as a non-admin user.	The Create Event button is hidden.	The button was not shown to non-admin users.	Pass
RBAC3	Request a user's own tickets.	Access is allowed and the ticket list is returned.	Users could access their own tickets correctly.	Pass
RBAC4	Request another user's tickets as an admin.	Access is allowed and the ticket list is returned.	Admin access was granted correctly.	Pass
RBAC5	Request another user's tickets as a non-admin.	Access is denied and an error is returned.	Unauthorized access was rejected correctly.	Pass

6.5 Waivers and Preference Submission Module

At the time of testing, no dedicated automated unit tests were implemented for waiver submission or preference submission as separate isolated modules. These behaviours were instead covered indirectly through broader event registration and ticketing workflows. Additional direct tests for waiver completion validation and user preference persistence may be added in future revisions.

7 Changes Due to Testing

[This section should highlight how feedback from the users and from the supervisor (when one exists) shaped the final product. In particular the feedback from the Rev 0 demo to the supervisor (or to potential users) should be highlighted. —SS]

8 Automated Testing

9 Trace to Requirements

Table [1](#) maps the selected tests to the main functional requirements. A checkmark indicates that the test provides evidence for the corresponding requirement.

Table 1: Traceability Between Tests and Functional Requirements

Test ID	FR1	FR2	FR3	FR4	FR5
ECM2					✓
ECM3					✓
ECM4					✓
ECM8		✓			✓
ECM9					✓
TP1	✓				
TP2	✓				
TP4	✓				
TP5	✓				
TP6	✓				
TP9	✓				
BTR1		✓			✓
BTR2		✓			✓
BTR3		✓			
BTR4		✓			
BTR5		✓			
RBAC1				✓	✓
RBAC2				✓	
RBAC3	✓			✓	
RBAC4	✓			✓	
RBAC5				✓	

10 Trace to Modules

Table 2 maps the selected tests to the major functional modules of the system. A checkmark indicates that the test belongs to or primarily validates that module.

Table 2: Traceability Between Tests and Modules

Test ID	TPPS	BTR	WPS	RBAC	ECM
ECM2					✓
ECM3					✓
ECM4					✓
ECM8		✓			✓
ECM9					✓
TP1	✓				
TP2	✓				
TP4	✓				
TP5	✓				
TP6	✓				
TP9	✓				
BTR1		✓			✓
BTR2		✓			✓
BTR3		✓			
BTR4		✓			
BTR5		✓			
RBAC1				✓	✓
RBAC2				✓	
RBAC3	✓			✓	
RBAC4	✓			✓	
RBAC5				✓	

11 Code Coverage Metrics

References

Appendix — Reflection

The information in this section will be used to evaluate the team members on the graduate attribute of Reflection.

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. Which parts of this document stemmed from speaking to your client(s) or a proxy (e.g. your peers)? Which ones were not, and why?
4. In what ways was the Verification and Validation (VnV) Plan different from the activities that were actually conducted for VnV? If there were differences, what changes required the modification in the plan? Why did these changes occur? Would you be able to anticipate these changes in future projects? If there weren't any differences, how was your team able to clearly predict a feasible amount of effort and the right tasks needed to build the evidence that demonstrates the required quality? (It is expected that most teams will have had to deviate from their original VnV Plan.)